BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, DI KHAN

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2021

MATHEMATICS

Past Paper 2021 • Paper Code: DIK-SSC-X-2021 (A)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION – A

Time Allowed: 20 min | **Maximum Marks:** 15 | **Paper Code:** DIK-SSC-X-2021 (A)

Q1. Choose the correct option:

Q1. The solution of $\sqrt{(2x+3)} = x$ is _____

(A) -1 (B) 9

(C) 3 (D) Both A&B

Q2. The third proportional of x and y is _____

(A) y^2/x (B) x/y

(C) xy (D) None

Q3. The most frequent value in the data is called _____

(A) Mean (B) Median

(C) G.M (D) Mode

Q4. The measure of the semi-circle is_____

(A) 360° (B) 180°

(C) 90° (D) 270°

Q5. The measure of the external angle of hexagon is _____

(A) $\pi/3$ (B) $\pi/4$

(C) $\pi/6$ (D) None

Q6. A chord 40 units long is contained in a circle of radius 25. The distance of the chord from the centre of the circle is _____

(a) 15 units (b) 31.2 units

(c) 47.1 units (d) 50.1 units

Q7. Identify the item that does not have the same ratio as the other three.

(A) 30/45 (B) 4 to 6

(C) 2:3 (D) 3 to 2

Q8. For any two sets A and B which are subsets of ∪, then (A∩B)' = _____

(A) A'∩B' (B) A'∪B'

(C) A'∪B (D) A'B'

Q9. However many tangents can be drawn from the point outside the circle?

(A) 1 (B) 2

(C) 3 (D) 4

Q10. If an object is above the level of observer then the angle formed between the horizontal line of sight is called

(a) Angle of depression (b) Angle of elevation

(c) Abtuse angle (d) None of these

Q11. If "f" is a function from A to B, then "f" is onto function if:

(a) Range f = B (b) Range f ≠ A

(c) Dom f = A (d) Second element of all ordered pair contained in f is not repeated

Q12. What is the quotient when $x^3 - 8x^2 + 16x - 5$ is divided by $x - 5$?

(a) $x^2 - x + 5$ (b) $x^2 - 3x + 2$

(c) $x^2 - 3x + 1$ (d) $x^2 - 13x + 49 + 240/(x+5)$

Q13. What is the alternando property of the following if a/b = c/d

(a) (a−b)/b = (c−d)/d (b) a/(a+b) = c/(c+d)

(c) a/c = b/d (d) b/a = d/c

Q14. The quadratic equation $(x-2)^2 = 0$ has solution_____

(a) One real solution (b) Two real solutions

(c) No real solution (d) None of these

Q15. The sum of the cube roots of unity is _____

(A) 1 (B) −1

(C) 2 (D) 0

SECTION – B

Time Allowed: 2:40 Hrs | **Maximum Marks:** 36 | **Paper Code:** DIK-SSC-X-2021 (A)

Q2. Attempt any 9 questions of the following:

(i) Solve the equation $x + 2 + 1/(x+2) = 3/2$

(ii) Examine the nature of the roots of the equation $3x^2 + 2x + 1 = 0$

(iii) If $x + 1$ and $x - 2$ are factors of polynomial $x^3 + ax^2 + bx + 2$, then using synthetic division. Find the values of a and b.

(iv) The area of a rectangular filed is 252sq.m. The length of its side is 9m longer than its width. Find its sides.

(v) Solve the equation: $((x-1)^2+(x+2)^2)/((x-1)^2-(x+2)^2) = -17/8$

(vi) The following is the distribution for the number of defective bulbs in 30 carton (packs). Find the standard deviation of defective bulbs. Number of defective bulbs: 0-2 2-4 4-6 6-8 8-10 Number of packs (f): 1 3 15 10 2

(vii) If A = {1,2,3,4}, B = {1,3,5} and R = {(x,y)/y<x} is a binary relation from A to B then write it in tabular form.

(viii) If 8 persons complete a work in 10 days, then how many days would 10 persons take to complete the same work?

(ix) Resolve into partial fraction $x/(x^2+4x-5)$

(x) Verify distributive property of union over inter section. A = {0, ±1,± 2, ± 3, ± 5,} B={-1, -2, -3, -4, -5} and C = {-1, -2, 3, 4}

(xi) Convert -210^0 to radians.

(xii) Prove that $(\sec x - \tan x)^2 = (1-\sin x)/(1+\sin x)$

SECTION – C

Maximum Marks: 24

Q3. The angle is a segment greats than a semi-circle is less than a right angle.

Q4. If two arcs of a circle are congruent then the corresponding chords are equal?

Q5. Perpendicular from the centre of the circle on a chord bisect it.

Q6. Inscribe a regular hexagon in a circle of radius 4cm.